Data Science

CSE558

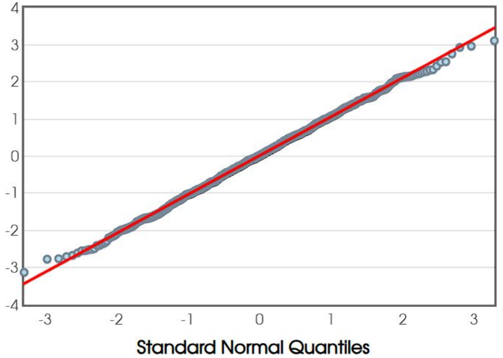
Overview

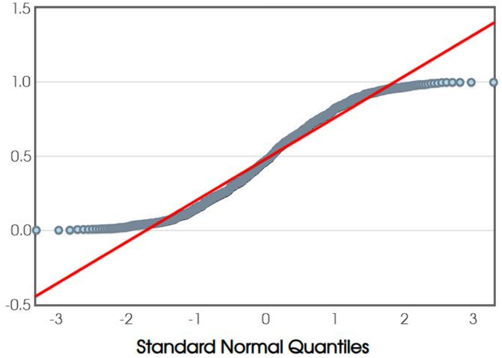
- Attribute types
- Statistical measures
  - Measure of central tendency & Dispersion
  - Correlations
- Plots
  - Quantile Plots

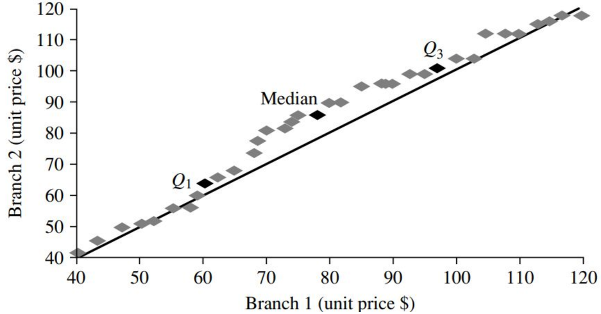
Overview

- Plots
- Probability Theory

Q-Plot



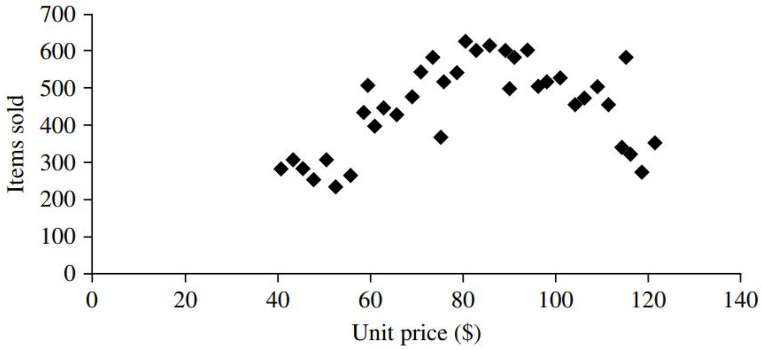




- Compare distributions of sampled data with theoretical distribution.
- Detect outliers.
- Compare distributions of two set of sampled data.

Scatter Plots

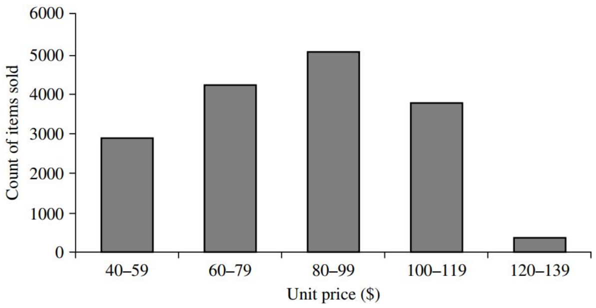
Plot high dimensional data



- Understand correlation in 2d data (+, -, linear, non-linear etc)
- Patterns – Clusters, Outliers
- Overplotting
- Correlation – Causation

Categorical Data

- Frequency measure for discrete
- Used for comparison



Bar graph

- Absolute measure

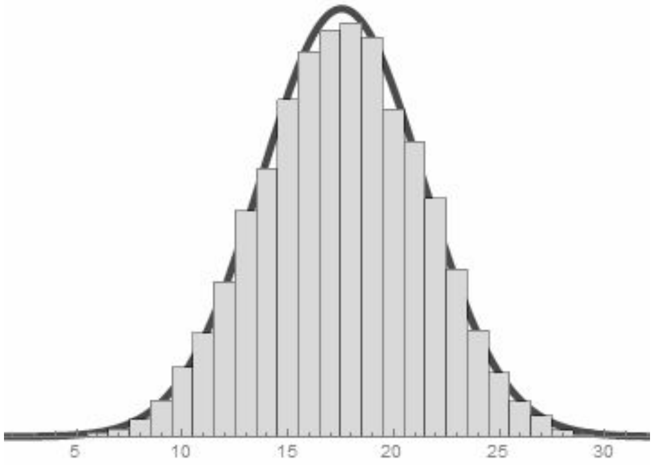
Pie Chart

- Relative Measure

Numeric Data

Histogram

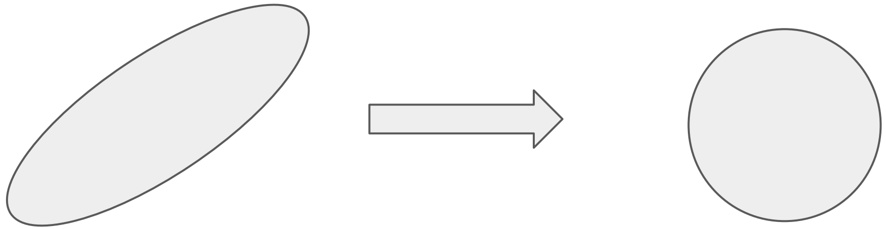
- Frequency measure Non-discrete values
- Distribution



Handling Outliers

1. Sort data
2. Visualize data – Box plots, Quantile plots, Scatter plots, histograms
3. Linear regression or Dimensionality reduction
4. Normalize data

$$\frac{x_i - \bar{x}}{\sigma}$$



Outliers even after normalize?

Handling Missing Values

Understand missed values

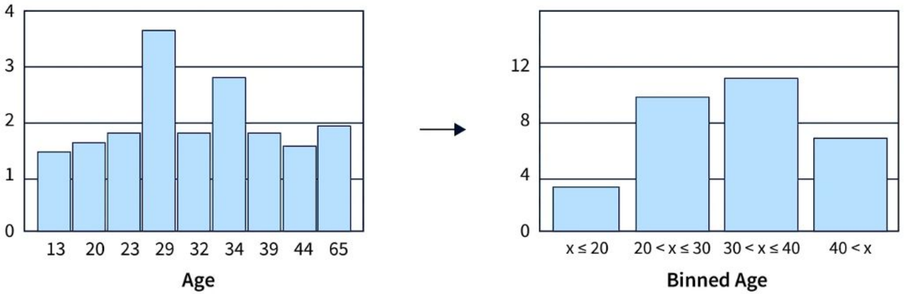
1. Deletion
2. Imputation
    a. Measure of central tendency
    b. Average kNN

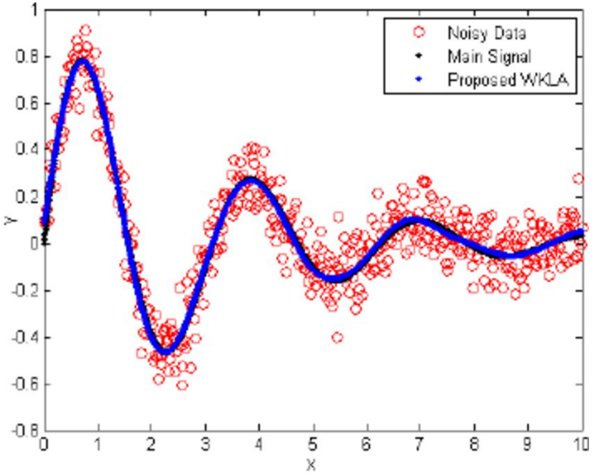
Handling Noise

Signal + noise.

Smoothening

- Binning
- Regression





Overview

- Plots
- Probability Theory

Probability

Probability Space: A triplet **(Ω, $\mathcal{F}$, $\mathbf{P}$)** that is used to define a random process.

- **Ω (Sample Space)**
  - $\Omega \neq \varnothing$.
  - Set of all outcomes of an experiment.
- **$\mathcal{F}$ (Event Space)**
  - σ-algebra
  - Set of outcomes in the sample space
- **$\mathbf{P}$ (Probability Measure)**
  - $\mathbf{P} : \mathcal{F} \rightarrow [0, 1]$

Probability Space - Example

Tossing a fair coin:

- **Ω: {H, T}**
- **$\mathcal{F}$: {{}, {H}, {T}, {H,T}}**
- **P: P({}) = 0, P({H}) = 1/2, P({T}) = 1/2, P({H,T}) = 1**

σ-algebra

Let $\mathbf{\Omega} \neq \emptyset$ be a set. Let $\mathcal{P}(\mathbf{\Omega})$ be its power set.

Then $\boldsymbol{\mathcal{F}} \subseteq \mathcal{P}(\mathbf{\Omega})$ is called a σ-algebra, if it satisfies:

1. Elements of $\boldsymbol{\mathcal{F}}$ are in sample Space, i.e., $\mathbf{\Omega} \in \boldsymbol{\mathcal{F}}$
2. $\boldsymbol{\mathcal{F}}$ is closed under complement:
   a. If $\mathbf{A} \in \boldsymbol{\mathcal{F}}$ then $\mathbf{A}^{\mathbf{c}} := \{ \mathbf{\Omega} \setminus \mathbf{A}\} \in \boldsymbol{\mathcal{F}}$
3. $\boldsymbol{\mathcal{F}}$ is closed under countable unions:
   a. If $\mathbf{A}_i \in \boldsymbol{\mathcal{F}}$ for i = 1,2,... then $\bigcup_{i=1}^{\infty} \mathbf{A}_i \in \boldsymbol{\mathcal{F}}$

Examples

Let $\Omega \neq \varnothing$ be some set.

- $\mathcal{F}_1 = \{\varnothing, \Omega\}$ is a σ-algebra commonly known as "trivial σ-algebra over Ω".
- $\mathcal{F}_2 = \mathcal{P}(\Omega)$ is a σ-algebra commonly known as "discrete σ-algebra over Ω".
- Let $A \subseteq \Omega$, then what are elements of the smallest σ-algebra that has **A**?
- $\mathcal{F}_3 = \{\varnothing, A, A^c, \Omega\}$.

Since $\mathcal{F}_3$ is generated by A so it is also denoted by $\sigma(\{A\})$.

Probability Measure

Let $\mathbf{\Omega} \neq \mathbf{Ø}$ be some set. Let $\mathcal{F} \subseteq \mathcal{P}(\mathbf{\Omega})$ be a σ-algebra.

Then a function $\mathbf{P} : \mathcal{F} \to [0, 1]$ is called a probability measure, if it satisfies:

1. The measure of $\mathbf{\Omega}$ is equal to one: $\mathbf{P}(\mathbf{\Omega}) = 1$
2. $\mathbf{P}$ is σ-additive: If $\{\mathbf{A}_i\}_{i=1}^{\infty} \subseteq \mathcal{F}$ is a countable collection of pairwise disjoint sets, i.e., $\mathbf{A_i} \cap \mathbf{A_j} = \mathbf{Ø}$ for every pair $\{\mathbf{A_i}, \mathbf{A_j}\}$ then

$$\mathbf{P}\left(\bigcup_{i=1}^{\infty} \mathbf{A}_i\right) = \sum_{i=1}^{\infty} \mathbf{P}(\mathbf{A}_i)$$

Example

Let $\mathbf{\Omega} = \{1, 2, 3\}$ and $\mathcal{F} = \mathcal{P}(\mathbf{\Omega})$.

Define $\mathbf{P} : \mathcal{F} \rightarrow [0, 1]$: $\mathbf{P}(1) = 0.3$, $\mathbf{P}(2) = 0.3$, $\mathbf{P}(3) = 0.4$

$$\sum_{\mathbf{x} \in \Omega} \mathbf{P}(\mathbf{x}) = \mathbf{P}(1) + \mathbf{P}(2) + \mathbf{P}(3) = 1$$

So, $\mathbf{P}(\mathbf{A}) := \sum_{\mathbf{x} \in \mathbf{A}} \mathbf{P}(\mathbf{x})$ for all $\mathbf{A} \in \mathcal{P}(\mathbf{\Omega})$.

For $\mathbf{A} = \{1, 2\}$ we get, $\mathbf{P}(\{1, 2\}) = \mathbf{P}(1) + \mathbf{P}(2) = 0.6$

Hence, $\mathbf{P}$ is a probability measure on $\mathcal{F}$.

Example

Let $\mathbf{\Omega}$ = {True, False} and $\mathcal{F} = \mathcal{P}(\mathbf{\Omega})$.

Define $\mathbf{P} : \mathcal{F} \rightarrow [0, 1]$: $\mathbf{P}$(True) = 0.45, $\mathbf{P}$(False) = 0.45

Is $\mathbf{P}$ a probability measure on $\mathcal{F}$?

No!

$$\sum_{\mathbf{x} \in \Omega} \mathbf{P}(\mathbf{x}) = \mathbf{P}(\mathrm{True}) + \mathbf{P}(\mathrm{Flase}) = 0.9$$

Question

Let $\Omega \neq \emptyset$ be a finite set and $\mathcal{F} = \mathcal{P}(\Omega)$.

Define $\mathbf{P}(\mathbf{A}) = \dfrac{|\mathbf{A}|}{|\Omega|}$ for every $\mathbf{A} \in \mathcal{F}$.

Prove or disprove that $\mathbf{P}$ a probability measure on $\mathcal{F}$.

Example

Consider a mcq quiz of 3 questions. Each question has 4 options, with a possibility of multiple correct answers.

- What is the sample space of possible correct and incorrect answers.
- To, pass if all the answers needs to be correct then what is the σ-algebra?
- If at least two answers needs to be correct then what is σ-algebra?
- If only one correct answers in each questions then what is the probability measure?

Conditional Probability

What is the probability that event $\mathbf{A}$ will occur given that event $\mathbf{B}$ has occurred?

$$\mathbf{P}(\mathbf{A}|\mathbf{B}) = \frac{\mathbf{P}(\mathbf{A} \cap \mathbf{B})}{\mathbf{P}(\mathbf{B})}$$

Note: The conditional probability is well-defined only if $\mathbf{P}(\mathbf{B}) > 0$.

Example

1. Let us toss two fair coins. What is the probability that the first toss is head given we get at least one head from the two toss?

   Let event $\mathbf{A}$: First toss is head.

   Let event $\mathbf{B}$: At least one head.

$$\mathbf{P}(\mathbf{A}|\mathbf{B}) = \frac{\mathbf{P}\left(\mathbf{A} \cap \mathbf{B}\right)}{\mathbf{P}(\mathbf{B})} = \frac{2/4}{3/4} = \frac{2}{3}$$

Example

2. Let us toss two fair coins. What is the probability that you get at least one head given that the first toss is head?

   Let event **A**: At least one head.

   Let event **B**: First toss is head.

   $\mathbf{P}(\mathbf{A}|\mathbf{B}) = 1$

3. Let us toss two fair coins. What is the probability that the second toss is heads given that the first toss was also heads?

Law of Total Probability

Let $\mathbf{\Omega} \neq \mathbf{Ø}$ be a sample space. Let $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_n$ be mutually disjoint events such that,

$\bigcup_{i=1}^{n} \mathbf{A}_i = \mathbf{\Omega}$. Let $\mathbf{B}$ be some event in $\mathbf{\Omega}$. Then, $\mathbf{P}(\mathbf{B}) = \sum_{i=1}^{n} \mathbf{P}(\mathbf{B}|\mathbf{A}_i) \cdot \mathbf{P}(\mathbf{A}_i)$

Since, $\mathbf{A}_i$ for $i = 1, 2, \ldots, n$ are disjoints and covers the entire $\mathbf{\Omega}$, so $\mathbf{B} \cap \mathbf{A}_i$ are also disjoints. So

$$\mathbf{P}(\mathbf{B}) = \sum_{i=1}^{n} \mathbf{P}(\mathbf{B} \cap \mathbf{A}_i) = \sum_{i=1}^{n} \mathbf{P}(\mathbf{B}|\mathbf{A}_i) \cdot \mathbf{P}(\mathbf{A}_i)$$

Bayes Rule?

Random Variables

Let $\mathbf{\Omega} \neq \mathbf{Ø}$ be a sample space.

Then a function $\mathbf{X} : \mathbf{\Omega} \to$ R is called a random variable,

Discrete random variable assigns values from a finite set or values from a countably infinite set.

Let $\mathbf{\Omega} \neq \mathbf{Ø}$ be a sample space and $\mathbf{X} : \mathbf{\Omega} \to$ R be a random variable,

Let $\mathbf{P} : \mathbf{\Omega} \to [0, 1]$ be a probability measure. Then, the **expectation** of $\mathbf{X}$ is,

$$\mathbf{E}[\mathbf{X}] = \sum_{i \in \Omega} \mathbf{X}(i) \cdot \mathbf{P}(i)$$

The expectation is finite if the summation converges, else it is infinite.

For any constant $\mathbf{c}$, $\mathbf{E}[\mathbf{c} \cdot \mathbf{X}] = \mathbf{c} \cdot \mathbf{E}[\mathbf{X}]$

Example

Tossing a fair coin.

What is the expectation of a fair coin being tossed?

Define a random variable that counts number of heads.

$$\mathbf{X} = \begin{cases} 1, & \text{if heads} \\ 0, & \text{if tails} \end{cases}$$

We know $\mathbf{P}$(heads) = 0.5 and $\mathbf{P}$(tails) = 0.5

So, $\mathbf{E}[\mathbf{X}]$ = $\mathbf{X}$(heads) $\cdot$ $\mathbf{P}$(heads) + $\mathbf{X}$(tails) $\cdot$ $\mathbf{P}$(tails) = 1 $\cdot$ 0.5 + 0 $\cdot$ 0.5 = 0.5

Example

Tossing a fair coin.

$$\mathbf{X} = \begin{cases} 7, & \text{if heads} \\ -3, & \text{if tails} \end{cases}$$

We know $\mathbf{P}$(heads) = 0.5 and $\mathbf{P}$(tails) = 0.5

So, $\mathbf{E}[\mathbf{X}]$ = $\mathbf{X}$(heads) $\cdot$ $\mathbf{P}$(heads) + $\mathbf{X}$(tails) $\cdot$ $\mathbf{P}$(tails) = 7 $\cdot$ 0.5 + (-3) $\cdot$ 0.5 = 2

Random variable does not affect the probability distribution over sample space.

Hence, the expectation of a random variable is a constant!

Linearity of Expectation

For any finite collection of discrete random variables, X_1, X_2, …, X_n with finite expectations,

$$\mathbf{E}\left[\sum_{i=1}^{n} \mathbf{X}_i\right] = \sum_{i=1}^{n} \mathbf{E}[\mathbf{X}_i]$$

Proof: We prove for just two random variables, say $\mathbf{X}$ and $\mathbf{Y}$

$$\mathbf{E}\left[\mathbf{X} + \mathbf{Y}\right] = \sum_i \sum_j (i + j) \cdot \mathbf{P}(\mathbf{X} == i \cap \mathbf{Y} == j)$$

$$= \sum_i i \sum_j \mathbf{P}(\mathbf{X} == i \cap \mathbf{Y} == j) + \sum_j j \sum_i \mathbf{P}(\mathbf{X} == i \cap \mathbf{Y} == j)$$

$$= \sum_i i \cdot \mathbf{P}(\mathbf{X} == i) + \sum_j j \cdot \mathbf{P}(\mathbf{Y} == j) \quad = \mathbf{E}[\mathbf{X}] + \mathbf{E}[\mathbf{Y}]$$

Bernoulli Random Variable

Let you run an experiment that succeed with probability ***p*** and fails with probability ***1-p***. A Bernoulli random variable $\mathbf{X}$ measures success or failure of the experiment,

$$\mathbf{X} = \begin{cases} 1, & \text{if the experiment succeeds} \\ 0, & \text{otherwise} \end{cases}$$

The random variable X is also called indicator random variable,

The expectation of the random variable is $\mathbf{E}[\mathbf{X}]$ = ***p***.

One event with success and failure!

Binomial Random Variable

A binomial random variable $\mathbf{X}$ with parameters **n** and ***p***, is denoted by $\mathbf{B}(\boldsymbol{n}, \boldsymbol{p})$. It takes non negative integer values and defines the probability distribution on $\boldsymbol{i}$ = 0, 1, …, $\boldsymbol{n}$.

$$\mathbf{P}(\mathbf{X} == i) = \binom{n}{i} p^i \cdot (1-p)^{n-i}$$

Multiple events with success & failure!

It is aggregation of Bernoulli random variables.

$\mathbf{E}[\mathbf{X}]$?

Geometric Random Variable

A geometric random variable $\mathbf{X}$ with parameter $\boldsymbol{p}$ takes positive integer values.
The probability distribution on **i** = 1,2, … is,

$$\mathbf{P}(\mathbf{X} == i) = (1-p)^{i-1} \cdot p$$

$\mathbf{E}[\mathbf{X}]$: # roll a die until you get 6.

$$\mathbf{E}[\mathbf{X}] = 1p + 2(1-p)p + 3(1-p)^2 p + 4(1-p)^3 p + \ldots$$

$$-\left((1-p)\mathbf{E}[\mathbf{X}] = 1(1-p)p + 2(1-p)^2 p + 3(1-p)^3 p + \ldots\right)$$

$$p\mathbf{E}[\mathbf{X}] = p + (1-p)p + (1-p)^2 p + \ldots$$

$$\mathbf{E}[\mathbf{X}] = 1 + (1-p) + (1-p)^2 + \ldots = \frac{1}{1-(1-p)} = \frac{1}{p}$$

Number of trials before having success.